University of California Riverside
Electrical and Computer Engineering Department

Electromagnetics I

Lecture 7: EM waves – mathematical description

Instructor: Dr. Alexander Khitun
Electrical and Computer Engineering Department
Email: akhitun@ece.ucr.edu

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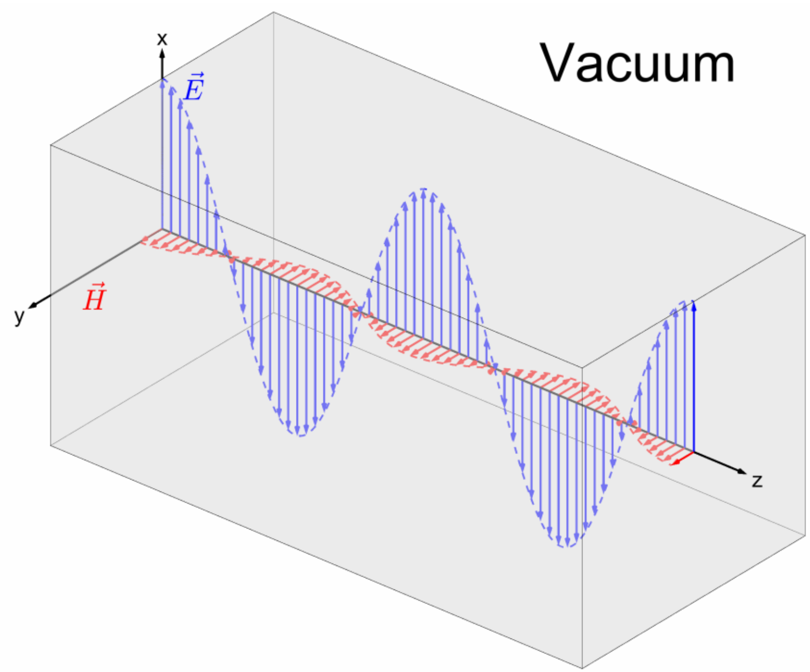
Outline

- EM time-domain representation
- EM phasor representation
- Intrinsic wave impedance
- Polarization
- Poynting's theorem
- Summary
- Examples



Chapter 2

Manipulation of Maxwell's equations leads to the following plane **wave equations** for $\vec{E}$ and $\vec{B}$:

Equations on this slide are for waves propagating along x-direction.

$$\frac{\partial^2 E_y}{\partial x^2} = \mu_0 \varepsilon_0 \frac{\partial^2 E_y(x,t)}{\partial t^2} \qquad \frac{\partial^2 B_z}{\partial x^2} = \mu_0 \varepsilon_0 \frac{\partial^2 B_z(x,t)}{\partial t^2}$$

These equations have solutions:

$$E_y = E_{max} \sin(kx - \omega t)$$

$$B_z = B_{max} \sin(kx - \omega t)$$

E_{max} and B_{max} are the electric and magnetic field amplitudes

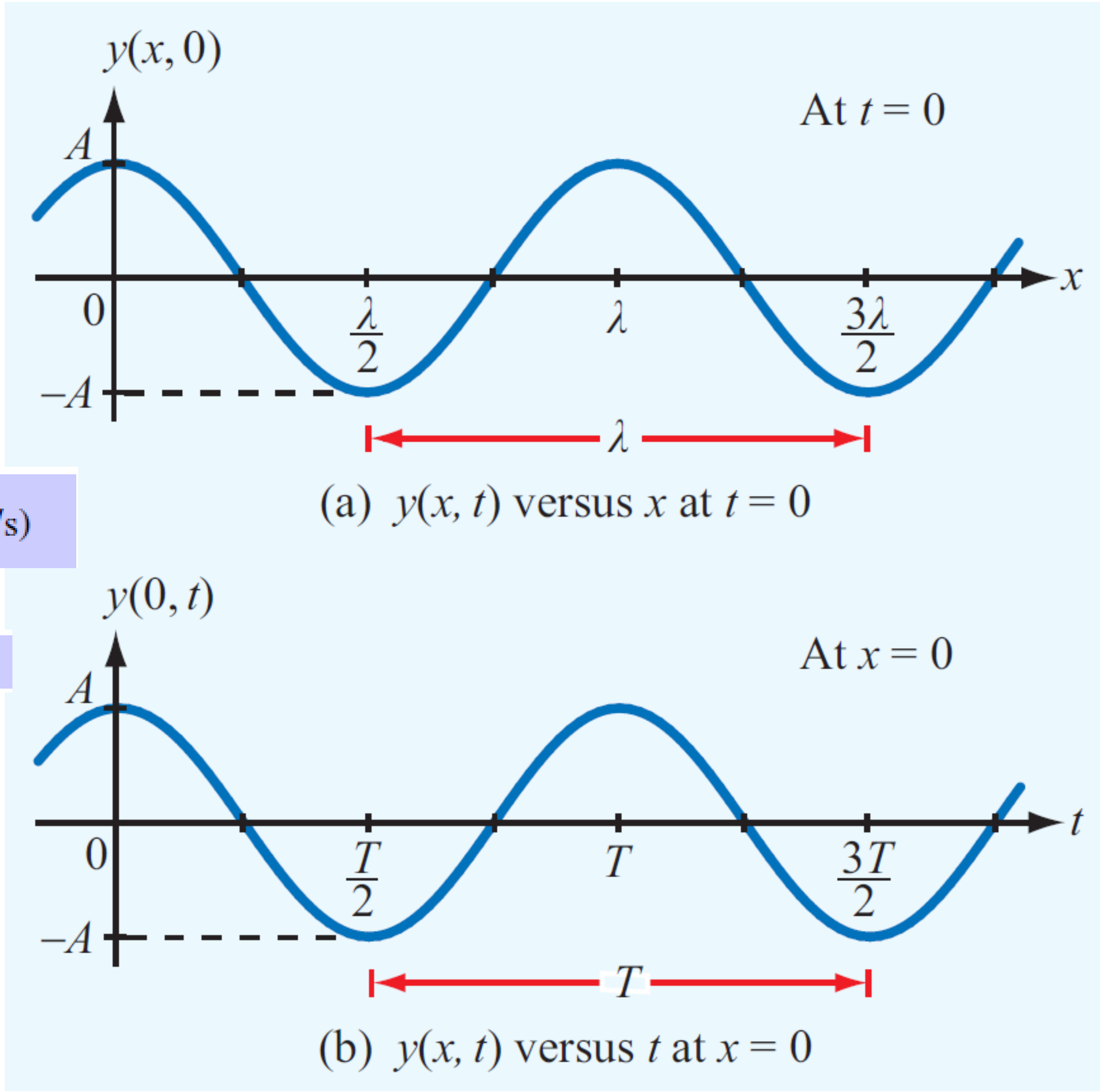
where $\boxed{k = \frac{2\pi}{\lambda},}$ $\boxed{\omega = 2\pi f,}$ and $\boxed{f\lambda = \frac{\omega}{k} = c.}$

You can verify this by direct substitution.

E_{max} and B_{max} in these notes are sometimes written by others as E_0 and B_0.

Sinusoidal wave

$$y(x,t) = A\cos(\frac{2\pi t}{T} - \frac{2\pi x}{\lambda} + \phi_0)$$



(a) $y(x, t)$ versus x at $t = 0$

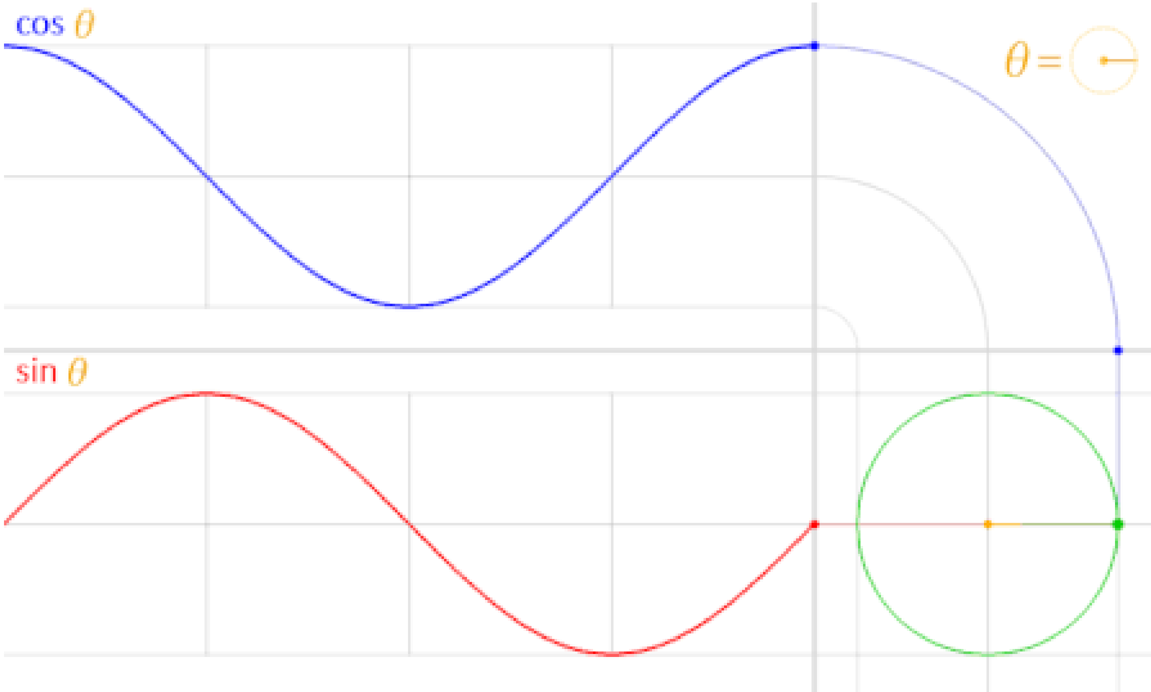
(b) $y(x, t)$ versus t at $x = 0$

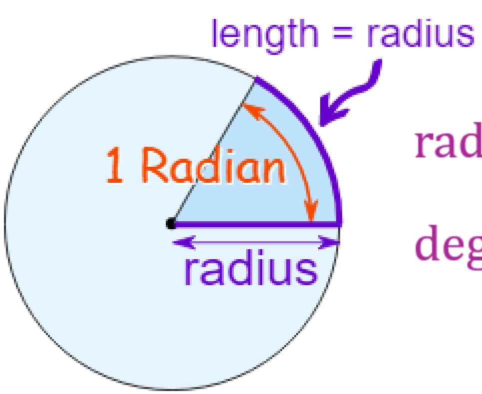
$$u_p = \frac{dx}{dt} = \frac{\lambda}{T} \quad \text{(m/s)}$$

$$u_p = f\lambda \quad \text{(m/s)}$$

Sine and Cosine functions

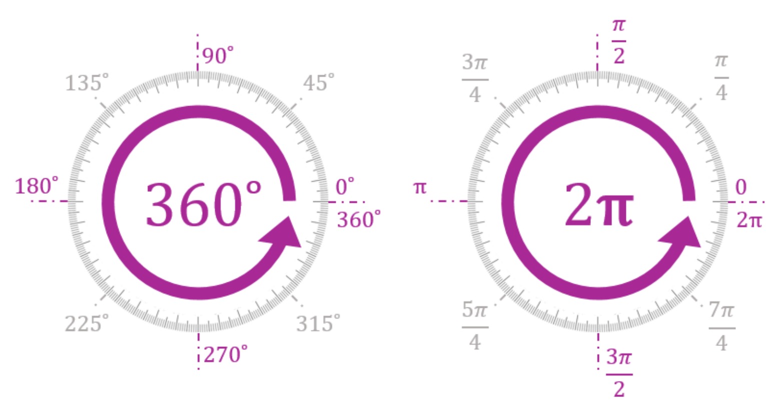
In mathematics, sine and cosine are trigonometric functions of an angle.





$$\text{radians} = \text{degrees} \times \pi / 180$$

$$\text{degrees} = \text{radians} \times 180 / \pi$$



Electro-magnetic waves

time-domain (instantaneous) expression

$$\vec{E}(z,t) = E_{max}\cos(\underbrace{2\pi \times ft}_{\omega} - \underbrace{\beta_0}_{\frac{2\pi}{\lambda}} z + \phi_0)\,\overrightarrow{a_x} \quad \text{V/m}$$

$$\vec{H}(z,t) = H_{max}\cos(2\pi \times ft - \beta_0 z + \phi_0)\,\overrightarrow{a_y} \quad \text{A/m}$$

- By defining

$$\text{radian frequency} = \omega = 2\pi f \quad \text{(rad/s)}$$

$$\text{wavenumber} = \beta = \frac{2\pi}{\lambda} \quad \text{(rad/m)}$$

Note: Radian frequency is also denoted *angular velocity* and wavenumber is also denoted *phase constant*

Reading the time-domain expression

$$f\lambda = \frac{\omega}{k} = c$$

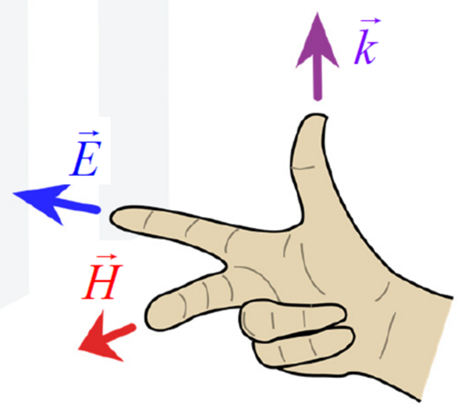
all electromagnetic waves propagate with the same speed in vacuum

Plane Waves in LHI Media are TEM

In linear, homogeneous, and isotropic (LHI) media, the electric field $\vec{E}$, magnetic field $\vec{H}$, and direction of the wave $\vec{k}$ are all perpendicular to each other. These are called transverse electromagnetic (TEM) waves.

$$\vec{E} \perp \vec{k} \perp \vec{H}$$

Further $\vec{E}$, $\vec{H}$, and $\vec{k}$ follow a right-hand rule.



Phasors

A **phasor** is a complex number that represents the amplitude and phase of a sinusoid.

A complex number z can be written in rectangular form as

$$z = x + jy \tag{9.14a}$$

$j = \sqrt{-1}$;

There are different ways to represent ***z***

$z = x + jy$	Rectangular form
$z = r\underline{/\phi}$	Polar form
$z = re^{j\phi}$	Exponential form

Representation of a Complex Number

$z = x + jy$ Rectangular form

$z = r\underline{/\phi}$ Polar form

$z = re^{j\phi}$ Exponential form

$$r = \sqrt{x^2 + y^2}, \qquad \phi = \tan^{-1}\frac{y}{x}$$

$$x = r\cos\phi, \qquad y = r\sin\phi$$

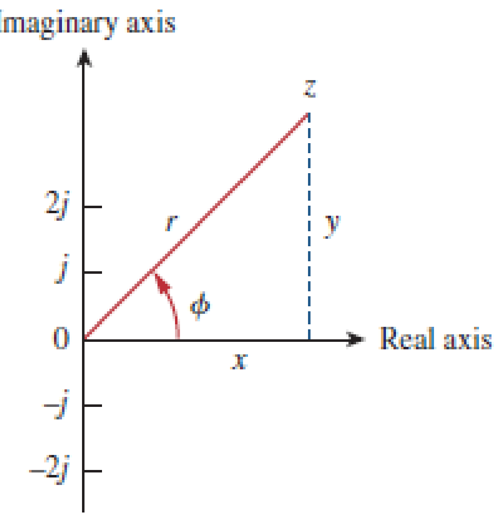


Figure 9.6
Representation of a complex number $z = x + jy = r\underline{/\phi}$.

$$\boxed{z = x + jy = r\underline{/\phi} = r(\cos\phi + j\sin\phi)}$$

Operations with Complex Numbers

$$z_1 = x_1 + jy_1 = r_1 \underline{/\phi_1}$$

$$z_2 = x_2 + jy_2 = r_2 \underline{/\phi_2}$$

Addition:

$$z_1 + z_2 = (x_1 + x_2) + j(y_1 + y_2)$$

Subtraction:

$$z_1 - z_2 = (x_1 - x_2) + j(y_1 - y_2)$$

Multiplication:

$$z_1 z_2 = r_1 r_2 \underline{/\phi_1 + \phi_2}$$

Division:

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} \underline{/\phi_1 - \phi_2}$$

Reciprocal:

$$\frac{1}{z} = \frac{1}{r} \underline{/-\phi}$$

Square Root:

$$\sqrt{z} = \sqrt{r} \underline{/\phi/2}$$

Complex Conjugate:

$$z^* = x - jy = r \underline{/-\phi} = re^{-j\phi}$$

Time Harmonic Fields

- Fields for which the time variation is sinusoidal are called time-harmonic fields.
- Plane waves are just one example of time-harmonic fields
- Maxwell's equations are significantly simplified in the case of harmonic fields

Time-Harmonic Fields and Complex Notation

- Basic idea:

If the time-variation of fields is known a-priori to be sinusoidal (i.e. the fields are known to be time-harmonic) then, in order to simplify the math, one may not carry around the time dependence explicitly in calculations

Some useful trigonometric Identities to recall before we start:

$$\cos(\theta) = \frac{e^{j\theta} + e^{-j\theta}}{2} \qquad e^{j\theta} = \cos(\theta) + j\sin(\theta)$$

$$\sin(\theta) = \frac{e^{j\theta} - e^{-j\theta}}{2j} \qquad e^{-j\theta} = \cos(\theta) - j\sin(\theta)$$

Expression for the E-field of a plane wave in complex notation:

$$\vec{E}(\vec{r},t) = \hat{n}\, E_o \cos\left(\omega t - \vec{k}.\vec{r}\right)$$

$$\vec{E}(\vec{r},t) = \hat{n}\, E_o \frac{e^{j\left(\omega t - \vec{k}.\vec{r}\right)} + e^{-j\left(\omega t - \vec{k}.\vec{r}\right)}}{2} = \mathrm{Re}\left[\hat{n}\, E_o e^{j\left(\omega t - \vec{k}.\vec{r}\right)}\right]$$

Time Harmonic Fields

$$E(r,t) \quad is \quad replaced \quad by \quad \hat{E}(r)e^{j\omega t}$$

$$\hat{E}(r) = E_0 e^{j\theta}$$

$$\nabla \cdot \epsilon_o(\hat{\mathbf{E}}(\mathbf{r}))e^{j\omega t} = \hat{\rho}\, e^{j\omega t}$$

$$\nabla \cdot (\hat{\mathbf{B}}(\mathbf{r})\, e^{j\omega t}) = 0$$

$$\nabla \times (\hat{\mathbf{E}}(\mathbf{r})\, e^{j\omega t}) = -j\omega(\hat{\mathbf{B}}(\mathbf{r})\, e^{j\omega t})$$

$$\nabla \times \left(\frac{\hat{\mathbf{B}}(\mathbf{r})\, e^{j\omega t}}{\mu_o}\right) = \hat{\mathbf{J}}(\mathbf{r})\, e^{j\omega t} + j\omega\epsilon_o(\hat{\mathbf{E}}(\mathbf{r})\, e^{j\omega t})$$

$$\nabla \cdot (\epsilon_o\, \hat{\mathbf{E}}(\mathbf{r})) = \hat{\rho}$$

$$\nabla \cdot \hat{\mathbf{B}}(\mathbf{r}) = 0$$

$$\nabla \times \hat{\mathbf{E}}(\mathbf{r}) = -j\omega\hat{\mathbf{B}}(\mathbf{r})$$

$$\nabla \times \frac{\hat{\mathbf{B}}(\mathbf{r})}{\mu_o} = \hat{\mathbf{J}}(\mathbf{r}) + j\omega\epsilon_o\, \hat{\mathbf{E}}(\mathbf{r})$$

$$\boxed{\begin{aligned} \mathbf{E}(\mathbf{r},t) &= Re(\hat{\mathbf{E}}(\mathbf{r})\, e^{j\omega t}) \\ \mathbf{B}(\mathbf{r},t) &= Re(\hat{\mathbf{B}}(\mathbf{r})\, e^{j\omega t}) \end{aligned}}$$

$$\begin{aligned} E_x(\mathbf{r},t) &= Re(E_o e^{j\theta} e^{j\omega t}) \\ &= E_o \cos(\omega t + \theta) \end{aligned}$$

Waves in Free Space

$$\boxed{\begin{aligned} &\nabla^2 \hat{E} + \omega^2 \mu_0 \epsilon_0 \hat{E} = 0 \\ &\nabla^2 \hat{B} + \omega^2 \mu_0 \epsilon_0 \hat{B} = 0 \end{aligned}}$$

$\hat{E} \equiv \hat{E}(\vec{r})$
$\hat{B} = \hat{B}(\vec{r})$

consider specific cases:

- uniform plane wave
- spherical wave
- cylindical wave

uniform plane wave (z)
$\Rightarrow$ no variation of the fields in the plane wave front: $\frac{\partial}{\partial x}(\hat{E}, \hat{B}) = \frac{\partial}{\partial y}(\hat{E}, \hat{B}) = 0$

$$\begin{cases} \bar{\nabla} \cdot \epsilon_0 \hat{E} = 0 \\ \bar{\nabla} \cdot \hat{B} = 0 \\ \bar{\nabla} \times \hat{B} = j\omega \epsilon_0 \mu_0 \hat{E} \\ \bar{\nabla} \times \hat{E} = -j\omega \hat{B} \end{cases}$$

$\Rightarrow$ (4th) Faraday's:

$$\begin{vmatrix} \bar{a}_x & \bar{a}_y & \bar{a}_z \\ 0 & 0 & \frac{\partial}{\partial z} \\ \hat{E}_x & \hat{E}_y & \hat{E}_z \end{vmatrix} = -j\omega(\hat{B}_x \vec{a}_x + \hat{B}_y \vec{a}_y + \hat{B}_z \vec{a}_z)$$

$$\begin{cases} -\frac{\partial \hat{E}_y}{\partial z} = -j\omega \hat{B}_x \\ \frac{\partial \hat{E}_x}{\partial z} = -j\omega \hat{B}_y \quad (*) \\ 0 = -j\omega \hat{B}_z \end{cases}
\qquad
\begin{cases} -\frac{\partial \hat{B}_y}{\partial z} = j\omega \epsilon_0 \mu_0 \hat{E}_x \quad (*) \\ \frac{\partial \hat{B}_x}{\partial z} = j\omega \epsilon_0 \mu_0 \hat{E}_y \\ 0 = j\omega \epsilon_0 \mu_0 \hat{E}_z \end{cases}$$

Uniform Plane Wave

2

- No electric or magnetic field components along z-direction : $\hat{E}_z = \hat{B}_z = 0$

- $\hat{E}_x, \hat{B}_y$ related to each other like $\hat{B}_x, \hat{E}_y$ $\Rightarrow$ consider $(\hat{E}_x, \hat{B}_y) \neq 0$

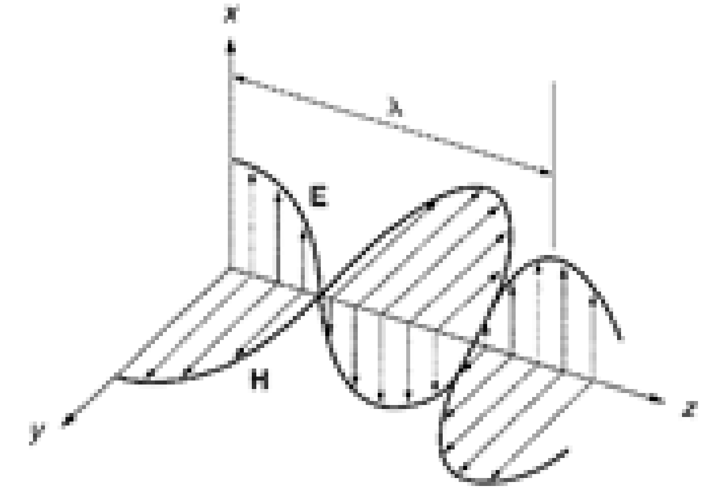
$$\left\{ \begin{aligned} \frac{\partial \hat{E}_x}{\partial z} &= -j\omega \hat{B}_y \\ -\frac{\partial \hat{B}_y}{\partial z} &= j\omega \epsilon_0 \mu_0 \hat{E}_x \end{aligned} \right. \qquad \frac{\partial^2 \hat{E}_x}{\partial z^2} = -j\omega \frac{\partial \hat{B}_y}{\partial z} = -\omega^2 \epsilon_0 \mu_0 \hat{E}_x$$

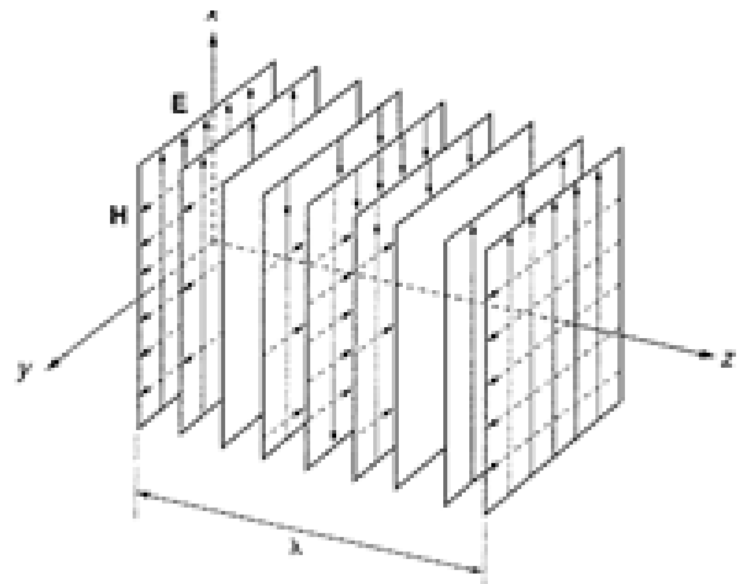
$$\Rightarrow \boxed{\frac{\partial^2 \hat{E}_x}{\partial z^2} + \omega^2 \mu_0 \epsilon_0 \hat{E}_x = 0}$$ scalar wave equation

General solution :

$$\boxed{\begin{aligned} &\hat{E}_x = \hat{C}_1 e^{-j\beta_0 z} + \hat{C}_2 e^{j\beta_0 z} \\ &\beta_0 = \omega\sqrt{\mu_0 \epsilon_0} \text{ - phase constant} \end{aligned}}$$ $\hat{C}_1$ and $\hat{C}_2$ - complex constants

$\rightarrow$ check





Propagating Waves

$e^{-j\beta_0 z}$ – change in phase corresponding propagation along $+z$ direction

$$\boxed{\hat{E}_x = \hat{E}_m^+ e^{-j\beta_0 z} + \hat{E}_m^- e^{j\beta_0 z}}$$

$\hat{E}_m^+$, $\hat{E}_m^-$ – complex or real ?

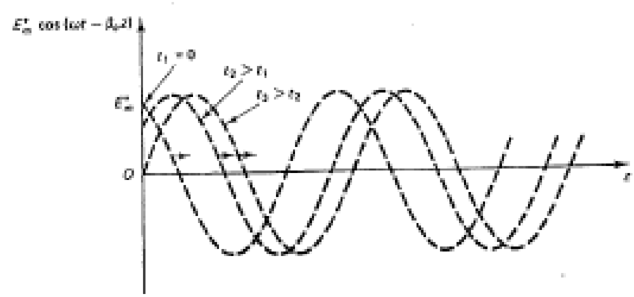
$$E_x(z,t) = Re\{\hat{E}_x e^{j\omega t}\} =$$

$$= Re\{E_m^+ e^{j(\omega t - \beta_0 z)} + E_m^- e^{j(\omega t + \beta_0 z)}\}$$

$$= E_m^+ \cos(\omega t - \beta_0 z) + E_m^- \cos(\omega t + \beta z)$$

traveling wave in $+z$ direction

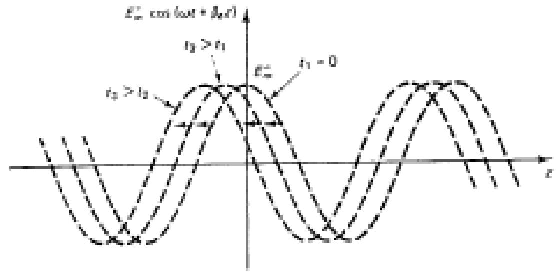
traveling wave in $-z$ direction



$$\omega t = \beta_0 z$$

$$z = \frac{\omega t}{\beta_0}$$

Propagating Wave (Cont.)



$$z = -\frac{\omega t}{\beta_0}$$

What about magnetic field?

$$\frac{\partial\left(E_m^+ e^{-j\beta_0 z}\right)}{\partial z} = -j\omega \hat{B}_y$$

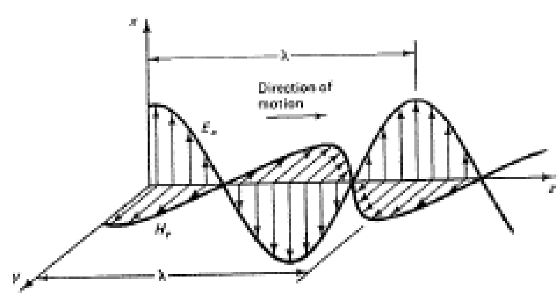
$$\Rightarrow \hat{B}_y = \frac{\beta_0}{\omega} E_m^+ e^{-j\beta_0 z} = \frac{\omega\sqrt{\mu_0 \epsilon_0}}{\omega} E_m^+ e^{-j\beta_0 z}$$

$$= \frac{E_m^+}{c} e^{-j\beta_0 z} = \frac{\hat{E}_x}{c}$$

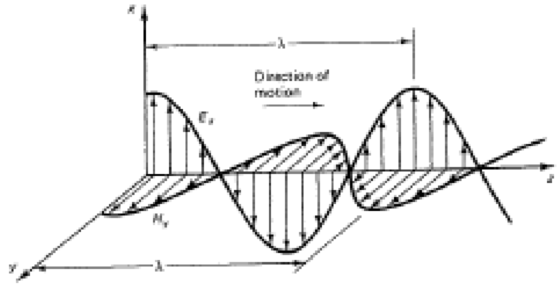
$$\boxed{c = \frac{1}{\sqrt{\epsilon_0 \mu_0}}}$$

velocity of light in vacuum

Uniform Plane Wave Propagation in Free Space



Propagating Waves



$$\hat{E}_x(z,t) = \hat{E}_m e^{-j\beta_0 z}$$

$$E_x(z,t) = E_m \cos(\omega t - \beta_0 z)$$

$$\hat{B}_y = \frac{\hat{E}_x}{c}$$

$$c = \frac{1}{\sqrt{\epsilon_0 \mu_0}} \approx 3 \cdot 10^8 \text{ m/s}$$

$\vec{E}$ and $\vec{B}$ are in phase

Technical/Engineering applications:

E [V/m]

$\hat{B}_m = \mu_0 \hat{H}_y$, $\hat{H}$ - magnetic field intensity

H [A/m] $\Rightarrow$ $\dfrac{E_x}{H_y} = \mu_0 c = \sqrt{\dfrac{\mu_0}{\epsilon_0}} = \eta_0 = 120\pi \approx$

$\approx 377\,\Omega$ - intrinsic wave impedance

$\boxed{\text{MKS units}}$

Wavelengh - distance z that the wave must travel to change the phase by 2π radians

$$\beta_0 \lambda = 2\pi \Rightarrow \boxed{\lambda = \frac{2\pi}{\beta_0} = \frac{c}{f}}$$

Propagating Waves

$e^{-j\beta_o z}$ - change in phase corresponding propagation along $+z$ direction

$$\boxed{\hat{E}_x = \hat{E}_m^+ e^{-j\beta_o z} + \hat{E}_m^- e^{j\beta_o z}}$$

$\hat{E}_m^+$, $\hat{E}_m^-$ - complex or real ?

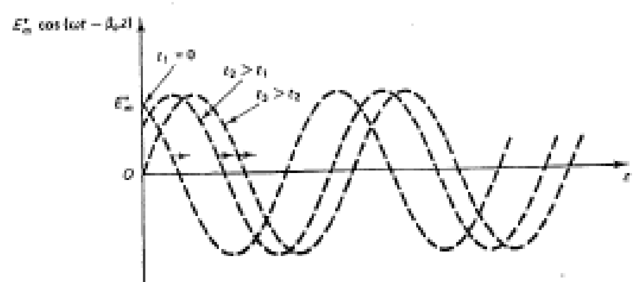
$$E_x(z,t) = Re\{\hat{E}_x e^{j\omega t}\} =$$

$$= Re\{E_m^+ e^{j(\omega t - \beta_o z)} + E_m^- e^{j(\omega t + \beta_o z)}\}$$

$$= E_m^+ \cos(\omega t - \beta_o z) + E_m^- \cos(\omega t + \beta z)$$

traveling wave in $+z$ direction

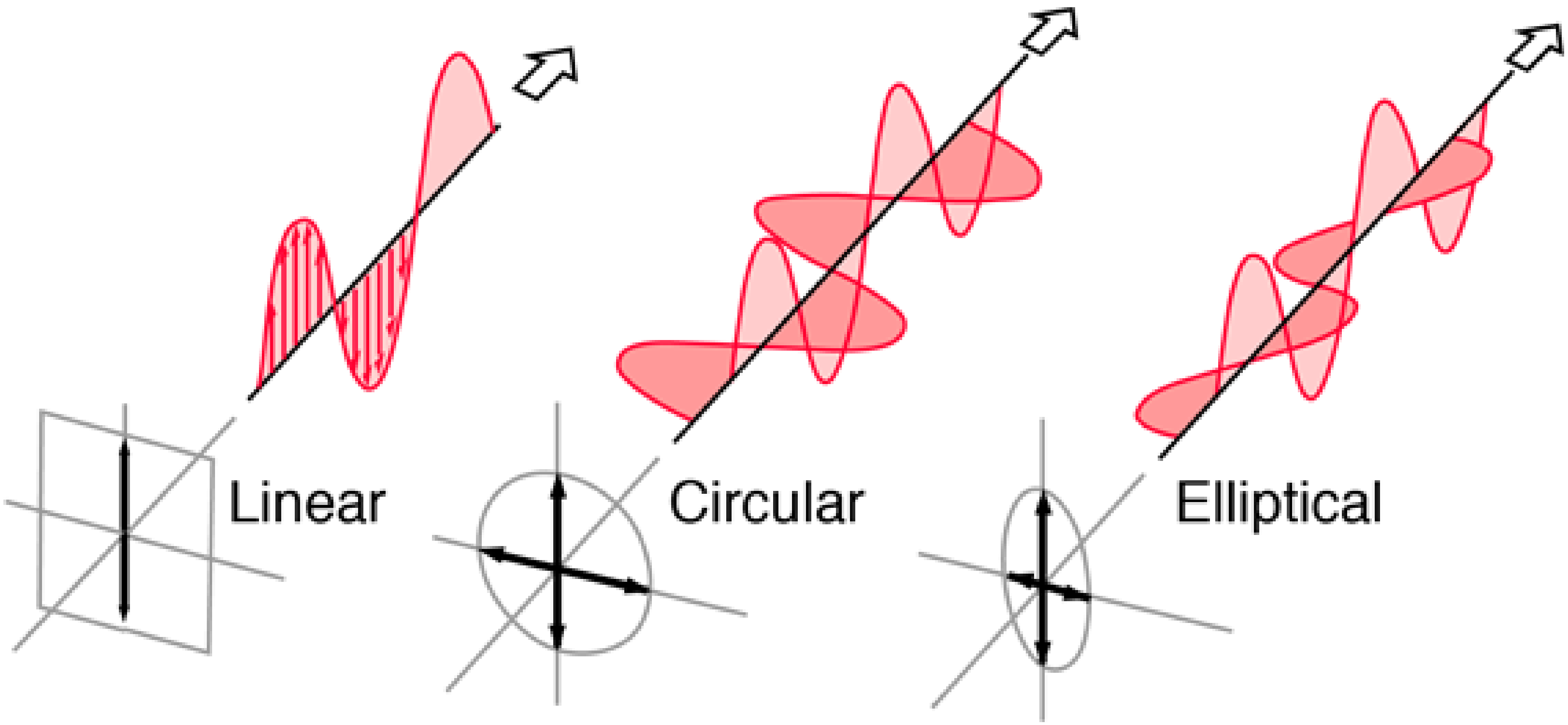
traveling wave in $-z$ direction



$$\omega t = \beta_o z$$

$$z = \frac{\omega t}{\beta_o}$$

Polarization of Plane Wave



Why is Polarization Important?

- Different polarizations can behave differently in a device
- Orthogonal polarizations will not interfere with each other
- Polarization becomes critical when analyzing devices on the scale of a wavelength
- Focusing properties of lenses are different
- Reflection/transmission can be different
- Frequency of resonators can be difference
- Cutoff conditions for filters, waveguides, etc.

Polarization of Plane Wave

$$\hat{\mathbf{E}} = (\hat{A}\,\mathbf{a}_x + \hat{B}\,\mathbf{a}_y)e^{-j\beta z}$$

where the amplitudes $\hat{A}$ and $\hat{B}$ may be complex, that is,

$$\hat{A} = |\hat{A}|e^{ja}, \qquad \hat{B} = |\hat{B}|e^{jb}$$

Elliptical

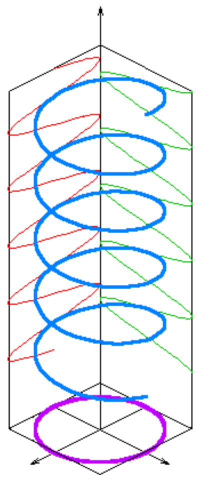
$a \neq b; |A| \neq |B| \neq 0$

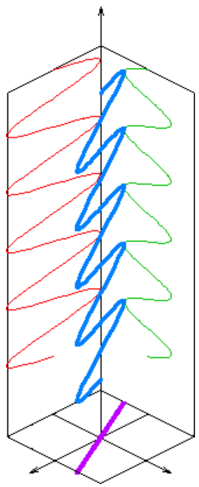
Circular

$|A| = |B|\ and\ |a - b| = \pi/2$

Linear

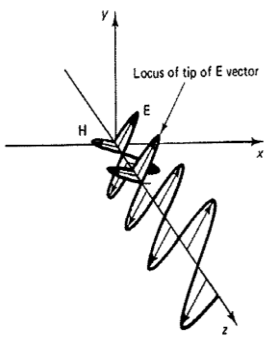
$a = b$





Polarization of Plane Waves ④

Polarization of Plane Waves



Propagation properties of a linearly polarized wave.

$(\hat{E}_x, \hat{H}_y)$

$(\hat{E}_y, \hat{H}_x)$

$$\hat{E} = (\hat{A}\vec{a}_x + \hat{B}\vec{a}_y)e^{-j\beta z}$$

$$\hat{A} = |A|e^{ja}$$

$$\hat{B} = |B|e^{jb}$$

① $a = b \Rightarrow \hat{E} = (|A|\vec{a}_x + |B|\vec{a}_y)e^{-j(\beta z - a)}$

real-time form: $\vec{E} = (|A|\vec{a}_x + |B|\vec{a}_y)\cos(\omega t - \beta z + a)$

$\vec{E}$ field is always in the plane perpendicular propagation direction

In case $a = b$, we have $\boxed{\text{linear polarization}}$

② $\hat{E} = \vec{a}_x|A|e^{j(a-\beta z)} + \vec{a}_y|B|e^{j(b-\beta z)}$

$$E_x = |A|\cdot\cos(\omega t + a - \beta z)$$

$$E_y = |B|\cos(\omega t + b - \beta z)$$

Polarization of Plane Waves ⑤

$a = 0,\ b = \pi/2 \Rightarrow$ locus of the end point of the $\vec{E}$ vector will trace out an ellipse

$$E_x(z,t) = |A|\cos(\omega t - \beta z)$$

$$E_y(z,t) = -|B|\sin(\omega t - \beta z)$$

$\boxed{\text{elliptical polarization}}$

ωt	Position
0	a
$\pi/4$	b
$\pi/2$	c
π	d
$5\pi/4$	e
$3\pi/2$	f
$7\pi/4$	g
2π	a

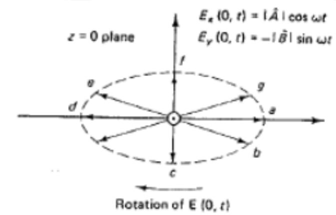


Figure 2.32 Rotation of the electric field vector associated with an elliptically polarized plane wave.

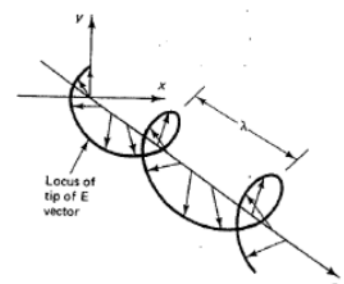


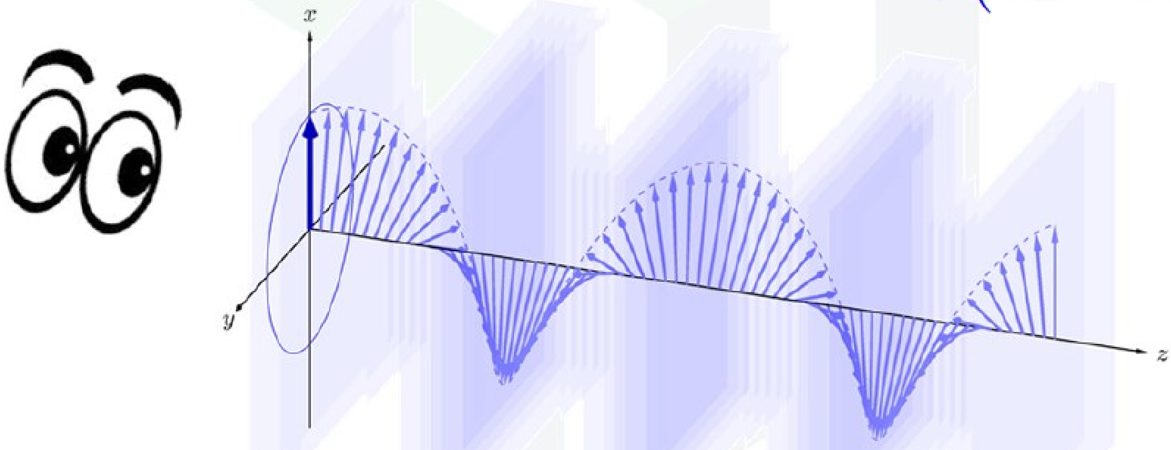
Figure 2.33 A circularly polarized wave in which the E field rotates as the wave advances.

$|A| = |B|,\ a - b = \pi/2 \Rightarrow$ $\boxed{\text{circular polarization}}$

Right Circular Polarization (RCP)

An electromagnetic wave has right circular polarization if the electric field rotates clockwise when viewed from behind.

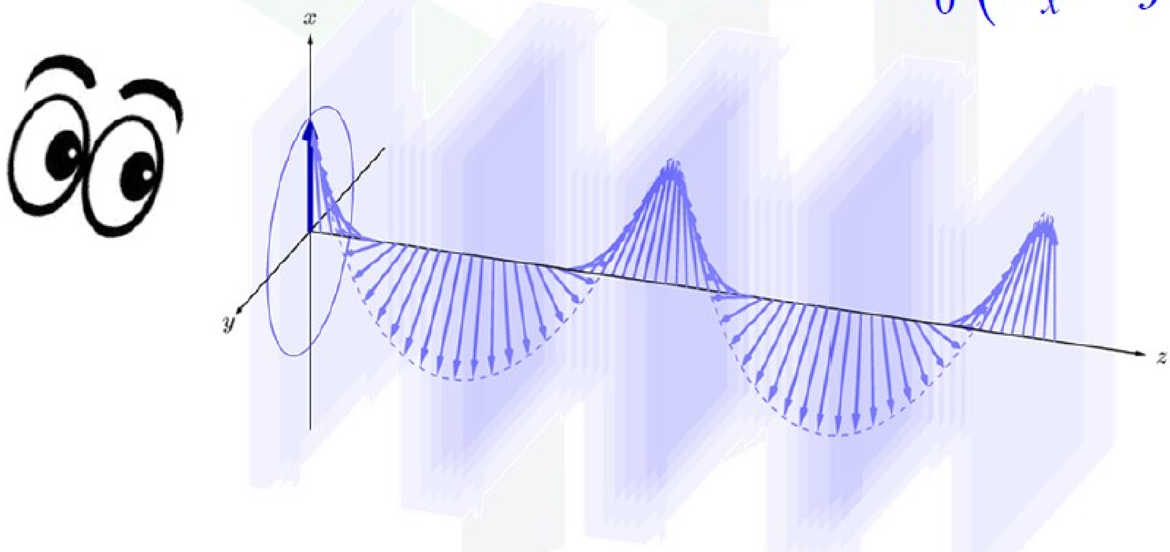
$$\vec{E} = E_0\left(\hat{a}_x + j\hat{a}_y\right)e^{-jkz}$$



Left Circular Polarization (LCP)

An electromagnetic wave has left circular polarization if the electric field rotates counterclockwise when viewed from behind.

$$\vec{E} = E_0\left(\hat{a}_x - j\hat{a}_y\right)e^{-jkz}$$



Poynting Theorem p 249

$\vec{P} = \vec{E} \times \vec{H}$ $\quad (V/m \cdot A/m = W/m^2)$

- power density vector along the direction of propagation (energy flow)

$$\boxed{\text{Poynting theorem deals with electromagnetic power balance (1884, English physicist)}}$$

Step 1: Vector identity: $\bar{\nabla} \cdot (\bar{E} \times \bar{H}) = \bar{H} \cdot (\bar{\nabla} \times \bar{E}) - \vec{E} \cdot (\bar{\nabla} \times \bar{H})$

Step 2: get $\bar{\nabla} \times \bar{E}$ & $\bar{\nabla} \times \bar{H}$ from Maxwell's equations

$$\bar{\nabla} \cdot (\bar{E} \times \bar{H}) = \bar{H}\left(-\frac{\partial \bar{B}}{\partial t}\right) - \bar{E}\left(\bar{J}_c + \frac{\partial \bar{D}}{\partial t}\right)$$

Step 3: rearranging term to see the physical meaning

$$\frac{\partial}{\partial t}\left(\frac{\bar{H} \cdot \bar{B}}{2}\right) = \frac{1}{2}\bar{H}\frac{\partial \bar{B}}{\partial t} + \frac{1}{2}\frac{\partial \bar{H}}{\partial t} \cdot \bar{B}$$

$B = \mu H$

$$\frac{\partial}{\partial t}\left(\frac{\bar{H} \cdot \bar{B}}{2}\right) = \frac{1}{2}\bar{H} \cdot \mu \frac{\partial \bar{H}}{\partial t} + \frac{1}{2}\frac{\partial \bar{H}}{\partial t} \cdot \mu \bar{H} =$$

$$= \bar{H}\frac{\partial \mu \bar{H}}{\partial t} = \bar{H} \cdot \frac{\partial \bar{B}}{\partial t}.$$ Similarly,

$$\frac{\partial}{\partial t}\left(\frac{\bar{E} \cdot \bar{D}}{2}\right) = \bar{E} \cdot \frac{\partial \bar{D}}{\partial t}$$

Poynting Theorem (Transmitter case)

- assumed tha EM field were generated outside V
- generator is inside V:

$$\bar{\nabla} \times \bar{H} = \bar{J}_g + \bar{J}_c + \frac{\partial \bar{D}}{\partial t}$$ $\Rightarrow$ additional "source term"

$$\underbrace{-\int_v \mathbf{J}_g \cdot \mathbf{E}\, dv}_{\text{Total power generated by the source}} = \underbrace{\frac{\partial}{\partial t}\left[\int_v \frac{\mathbf{E}\cdot\mathbf{D}}{2} dv + \int_v \frac{\mathbf{H}\cdot\mathbf{B}}{2} dv\right]}_{\text{Rate of increase of the electric and magnetic stored energies}} + \underbrace{\int_v \mathbf{E}\cdot\mathbf{J}_c\, dv}_{\text{Power dissipated}} + \underbrace{\oint_s \mathbf{E}\times\mathbf{H}\cdot ds}_{\text{Power density emanating from } s}$$

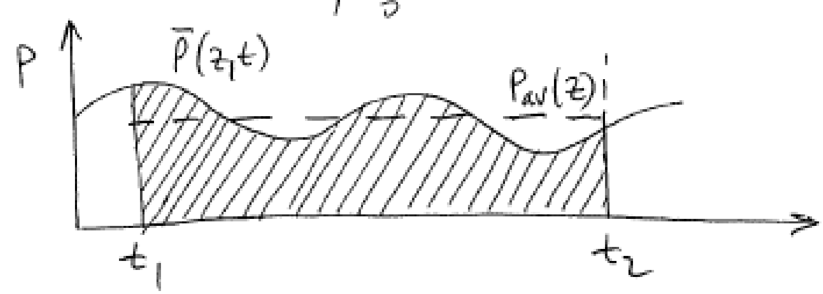
$\bar{P} = \bar{E} \times \bar{H}$ - instantaneous power density associated with propagation of EM wave

Practical Tasks: Time-average Poynting Vector

$$\vec{P}_{av}(z) = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} \vec{P}(z,t)\, dt$$

or for periodic functions:

$$\vec{P}_{av}(z) = \frac{1}{T} \int_0^T \bar{P}(z,t)\, dt$$



Summary

- EM waves can be described in time-domain and complex forms
- Units are important
- EM propagating in vacuum have the same speed (frequency-wavelength dependence)
- Wavefront – a locus of points with the same phase
- Polarization (elliptical, circular, and linear)
- Wave impedance: relates the magnitudes of electric and magnetic fields
- Poynting vector: units W/m^2, shows the direction of energy flow

EXAMPLE 2.23

The time domain (instantaneous) expression for the magnetic field intensity of a uniform plane wave propagating in the positive z direction in free space is given by

$$\mathbf{H}(z,t) = 4 \times 10^{-6} \cos(2\pi \times 10^7 t - \beta_o z)\,\mathbf{a}_y \text{ A/m}$$

1. Determine the phase constant β_o, and the wavelength λ.
2. Write a time domain expression for the electric field intensity $\mathbf{E}(z,t)$.

Solution

1. An important procedure for obtaining the wave parameters such as β_o, λ, and so forth is to compare the given expressions with those derived in our analysis. For example, from equation 2.62, we know that for a wave traveling along the positive z direction, the complex expression for the magnetic field intensity is given by

$$\hat{H}_y(z) = \frac{\hat{E}_x}{\eta_o} = \frac{\hat{E}_m^+}{\eta_o} e^{-j\beta_o z}$$

and the real-time form is, hence,

$$\mathbf{H}(z,t) = Re(\hat{H}_y(z) e^{j\omega t}) = Re\left(\frac{E_m^+}{\eta_o} e^{-j\beta_o z} e^{j\omega t}\right)$$

$$= \frac{E_m^+}{\eta_o} \cos(\omega t - \beta_o z)\,\mathbf{a}_y$$

Now comparing this expression with the given one for $\mathbf{H}$, it is clear that

$$\omega = 2\pi \times 10^7 \text{ rad/s}$$

and

$$f = \frac{\omega}{2\pi} = 10^7 = 10 \text{ MHz}$$

The wavelength $\lambda = c/f = 3 \times 10^8/10^7 = 30$ m. The phase constant β_o is obtained by noting from equation 2.64 that the phase velocity $v_p = c = \omega/\beta_o$.

$$\therefore \beta_o = \frac{\omega}{c} = \frac{2\pi \times 10^7}{3 \times 10^8} = 0.21 \text{ rad/m}$$

The phase constant may be alternatively obtained from equation 2.63,

$$\beta_o = \frac{2\pi}{\lambda} = \frac{2\pi}{\lambda} = 0.21 \text{ rad/m}$$

2. We know from the previous discussion that for a wave traveling along the positive z direction and with a magnetic field in the $\mathbf{a}_y$ direction, the electric field will be in the $\mathbf{a}_x$ direction. We also know that the amplitude of the electric field is related to that of the magnetic field by a constant ratio called intrinsic wave impedance η_o. Hence,

$$\begin{aligned} \mathbf{E}(z,t) &= \eta_o \times 4 \times 10^{-6} \cos(2\pi \times 10^7 t - \beta_o z)\,\mathbf{a}_x \text{ V/m} \\ &= 1.51 \times 10^{-3} \cos(2\pi \times 10^7 t - 0.21 z)\,\mathbf{a}_x \text{ V/m} \end{aligned}$$

EXAMPLE 2.24

A uniform plane wave is traveling in the positive z direction in free space. The amplitude of the electric field E_x is 100 V/m, and the wavelength is 25 cm.

1. Determine the frequency of the wave.
2. Write complete-complex and time-domain expressions for the electric and magnetic field vectors.

Solution

1. For a plane wave propagating in free space, the wavelength is related to the frequency in equation 2.63 by

$$\lambda = \frac{c}{f} \qquad \text{or} \qquad f = \frac{c}{\lambda}$$

   The frequency is then

$$f = \frac{3 \times 10^8}{0.25} = 12 \times 10^8 \text{ Hz}$$

$$= 1.2 \text{ GHz}$$

2. The complex expression for the electric field is

$$\bar{E}(z) = 100e^{-j\beta_o z}\,\mathbf{a}_x$$

To determine β_o, we use equation 2.63,

$$\beta_o = \frac{2\pi}{\lambda} = 8\pi$$

$$\therefore \hat{\mathbf{E}}(z) = 100e^{-j8\pi z}\,\mathbf{a}_x$$

The complex expression for the magnetic field intensity is

$$\hat{\mathbf{H}}(z) = \frac{100}{\eta_o}e^{-j8\pi z}\,\mathbf{a}_y$$

or

$$\hat{\mathbf{H}}(z) = \frac{100}{377}e^{-j8\pi z}\,\mathbf{a}_y$$

$$= 0.27e^{-j8\pi z}\,\mathbf{a}_y$$

The real-time forms of these fields are obtained from equation 2.59, hence,

$$\mathbf{E}(z,t) = 100\cos(24\pi \times 10^8 t - 8\pi z)\,\mathbf{a}_x$$

and

$$\mathbf{H}(z,t) = 0.27\cos(24\pi \times 10^8 t - 8\pi z)\,\mathbf{a}_y$$

Retrieval Session

Electric field of EM wave is given by:

$$\hat{E} = \widehat{E_x} + \widehat{E_y};$$

$$\widehat{E_x} = 3\ (V/m) \cdot e^{j\pi/3}$$

$$\widehat{E_y} = 7\ (V/m) \cdot e^{j\pi/3}$$

The wave polarizations:

a) Linear
b) Circular
c) Elliptical
d) Not possible to define

Retrieval Session

Electric field of EM wave is given by:

$$\hat{E} = \widehat{E_x} + \widehat{E_y};$$

$$\widehat{E_x} = 2\ (V/m) \cdot e^{j\pi/3}$$

$$\widehat{E_y} = 5\ (V/m) \cdot e^{j\pi/6}$$

The wave polarizations:

a) Linear
b) Circular
c) Elliptical
d) Not possible to define

Retrieval Session

Electric field of EM wave is given by:

$\hat{E} = \widehat{E_x} + \widehat{E_y};$

$\widehat{E_x} = 5\ (V/m) \cdot e^{j\pi/2}$

$\widehat{E_y} = 5\ (V/m) \cdot e^{j\pi}$

The wave polarizations:

a) Linear
b) Circular
c) Elliptical
d) Not possible to define